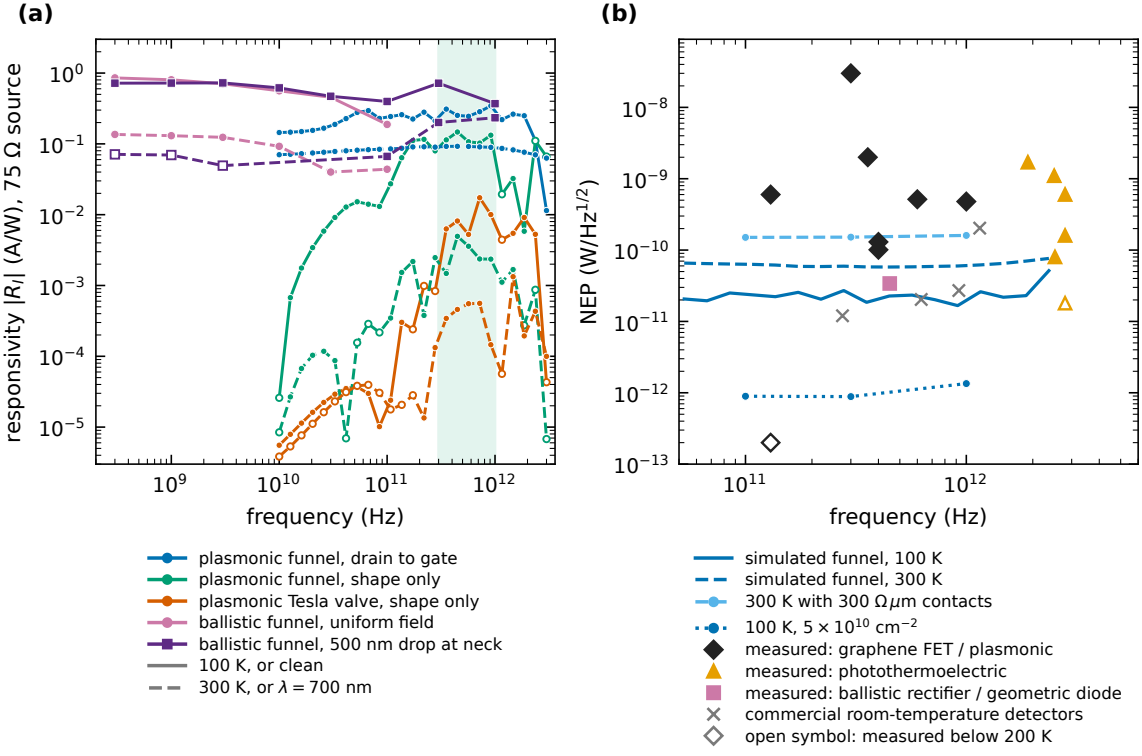
(a)
responsivity $|R_I|$ (A/W), 75 Ω source
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^9
10^{10}
10^{11}
10^{12}
frequency (Hz)
plasmonic funnel, drain to gate
plasmonic funnel, shape only
plasmonic Tesla valve, shape only
ballistic funnel, uniform field
ballistic funnel, 500 nm drop at neck
100 K, or clean
300 K, or $\lambda = 700$ nm
(b)
NEP (W/Hz$^{1/2}$)
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{11}
10^{12}
frequency (Hz)
simulated funnel, 100 K
simulated funnel, 300 K
300 K with 300 Ωμm contacts
100 K, 5×10^{10} cm^{-2}
measured: graphene FET / plasmonic
measured: photothermoelectric
measured: ballistic rectifier / geometric diode
commercial room-temperature detectors
open symbol: measured below 200 K